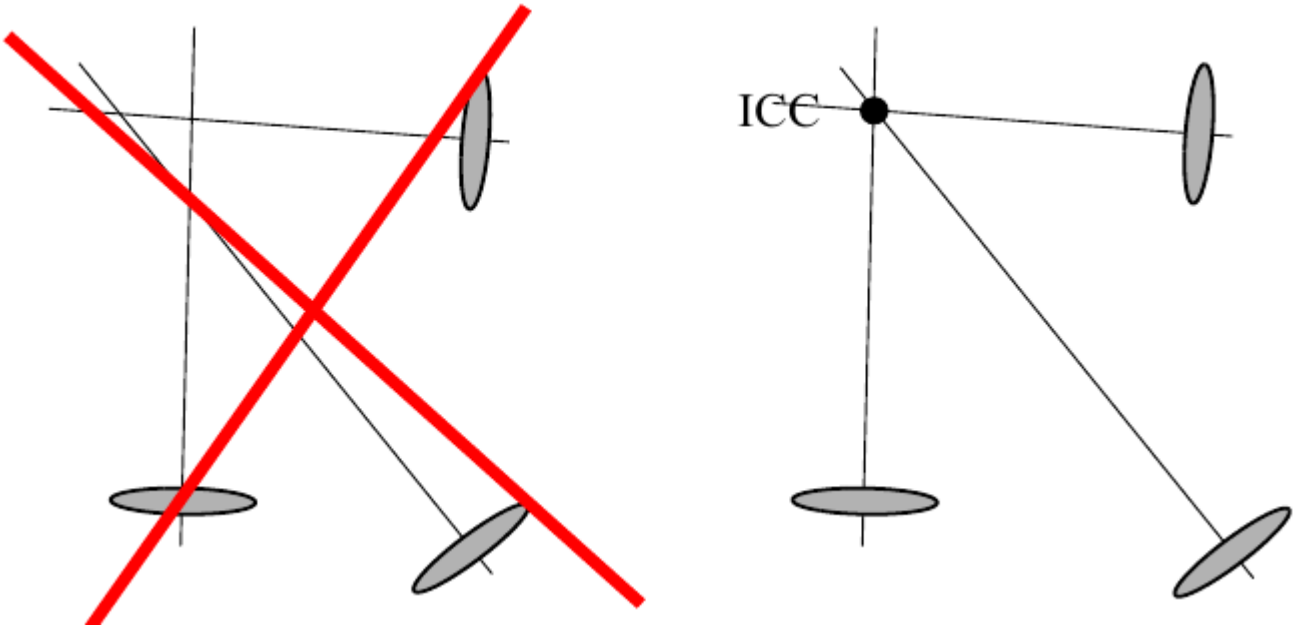
Ro04: Locomotion

- The "power of motion from place to place"
- Relevant for state estimation

Typical drives

- Differential
- Ackerman
- Synchronous
- Mecanum wheels

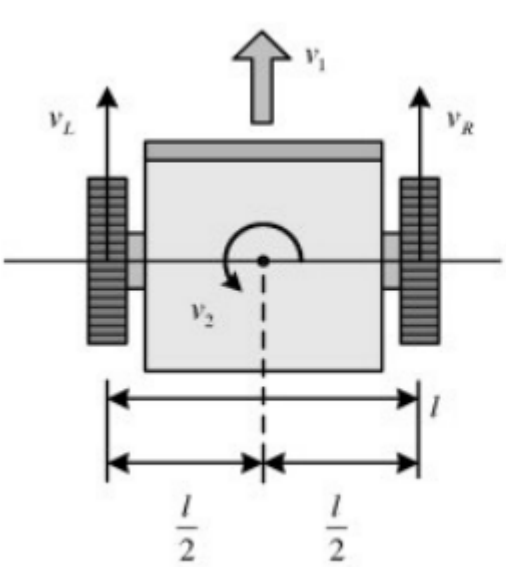
Instantaneous Center of Curvature

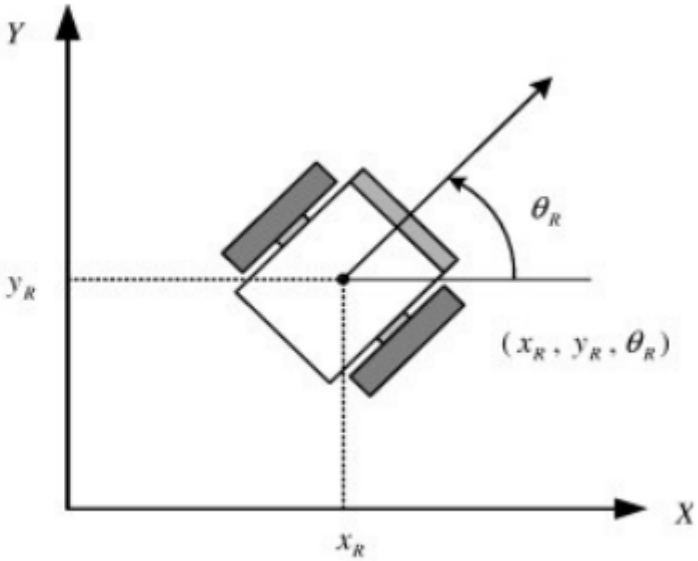
- For rolling motion to occur, the axes of all wheels must intersect in one point
- 

Basic Laws of Physics to Describe Wheeled Motion

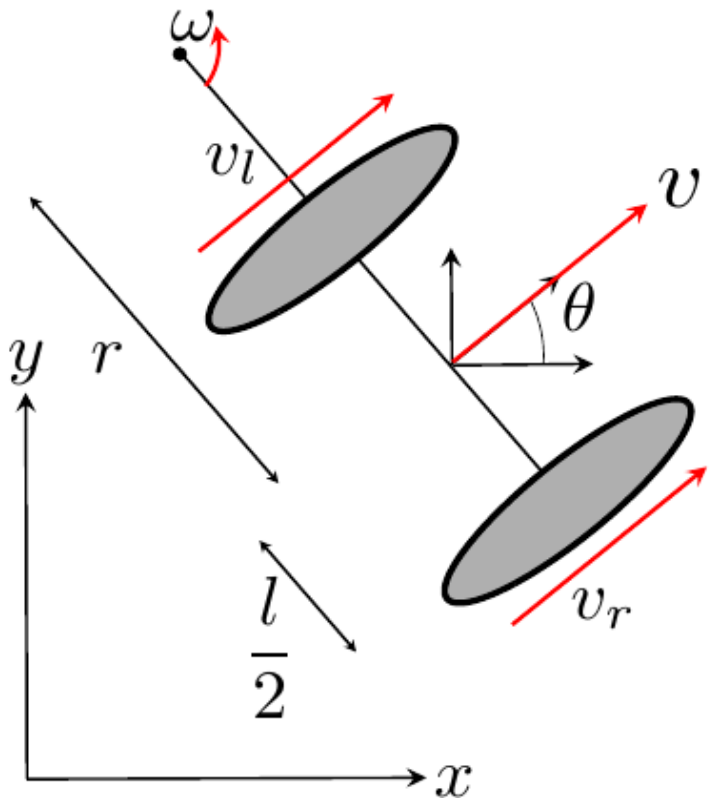
$$s = vt$$
$$\theta = \omega t$$
$$v = \omega r$$

Differential Drive Robot





- Unknowns:
  - translation velocity v
  - rotational velocity ω
  - radius of curve r

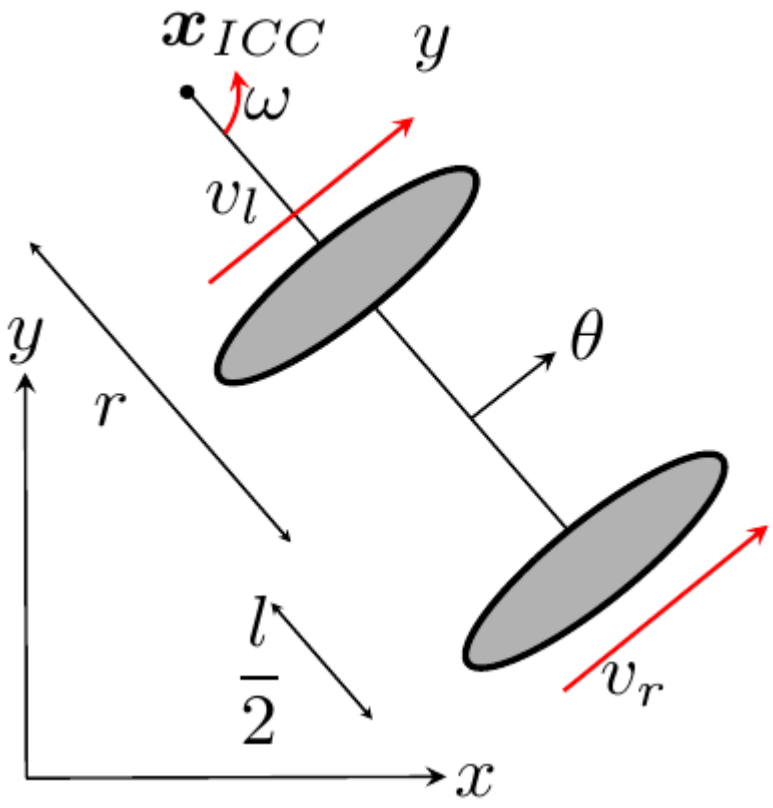


$$v_l = \omega\left(r - l/2\right)$$
$$v_r = \omega\left(r + l/2\right) \tag{1}$$

1. $\omega = \frac{v_r}{r+l/2} = \frac{v_l}{r+l/2}$
   - $r = \frac{l}{2}\frac{(v_r+v_l)}{(v_r-v_l)}$
2. $(v_r + v_l) = \omega(r + l/2 + r - l/2) = \omega 2r$
   - $\omega = \frac{(v_r+v_l)}{2r}$
   - $\omega = \frac{(v_r-v_l)}{l}$

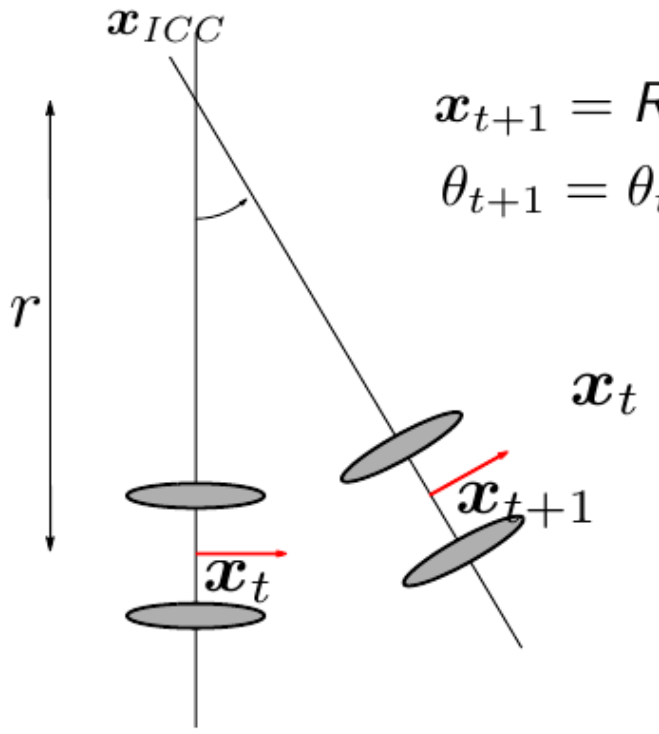
3. $v = \omega r$
   - $v = \frac{(v_r + v_l)}{2}$

ICC - Instantaneous Center of Curvature



$$\boldsymbol{x}_{ICC} = \begin{bmatrix} x - r\sin\theta \\ y + r\cos\theta \end{bmatrix}$$

Forward Kinematics

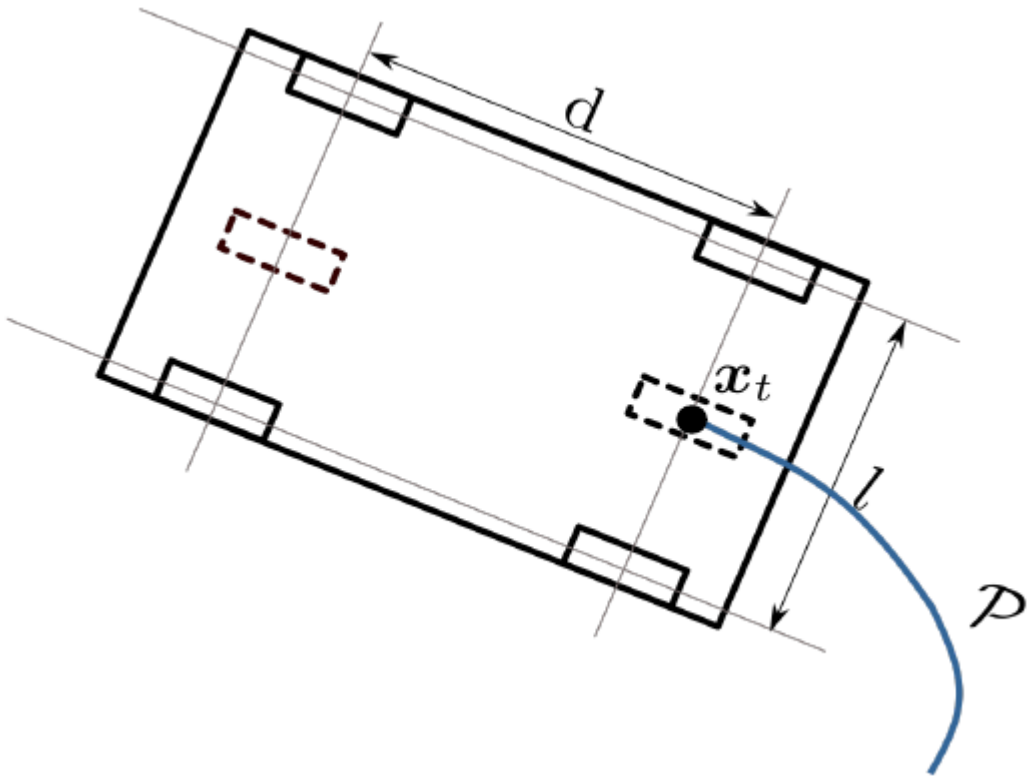


$$\boldsymbol{x}_{t+1} = R(\omega\,\Delta t)(\boldsymbol{x}_t - \boldsymbol{x}_{ICC}) + \boldsymbol{x}_{ICC}$$
$$\theta_{t+1} = \theta_t + \omega\,\Delta t$$

$$x_t = \int_0^t v(t')\cos\theta_{t'}\,dt'$$
$$y_t = \int_0^t v(t')\sin\theta_{t'}\,dt'$$
$$\theta_t = \int_0^t \omega(t')\,dt'$$

Ackermann Drive

- E.g. for modelling a car



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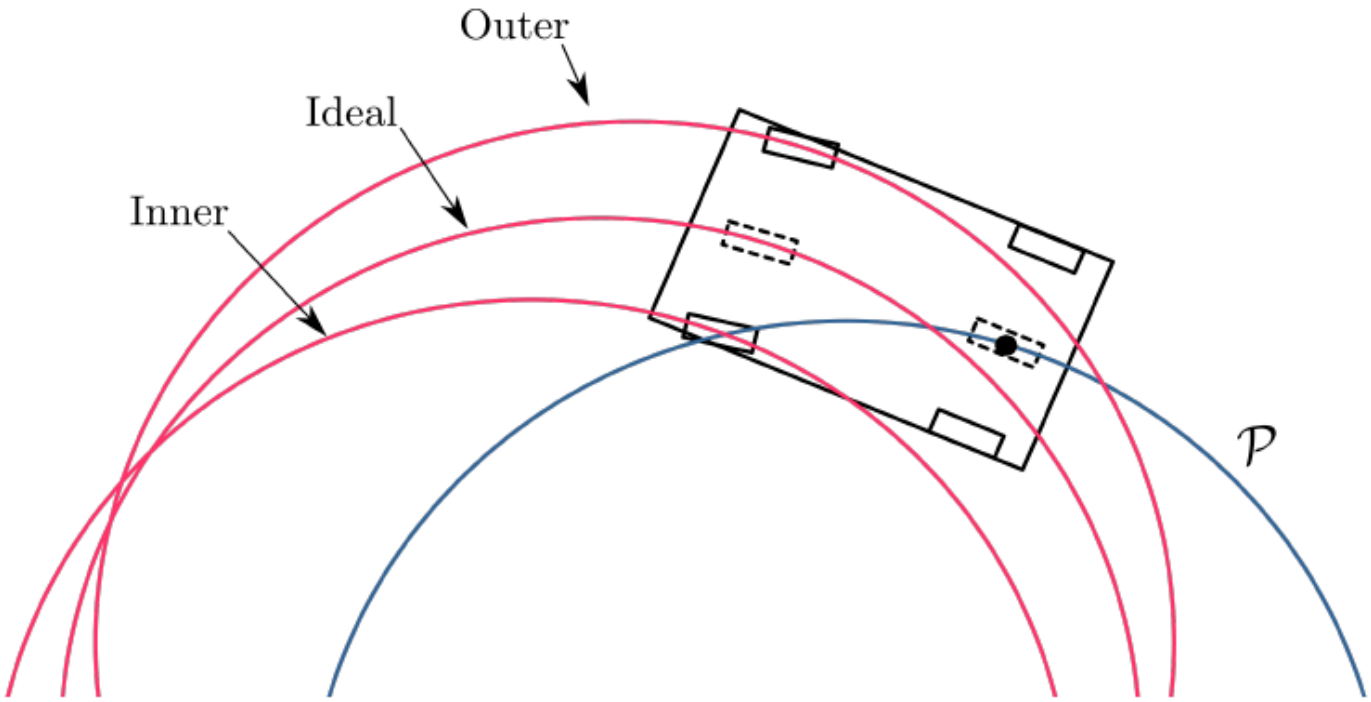
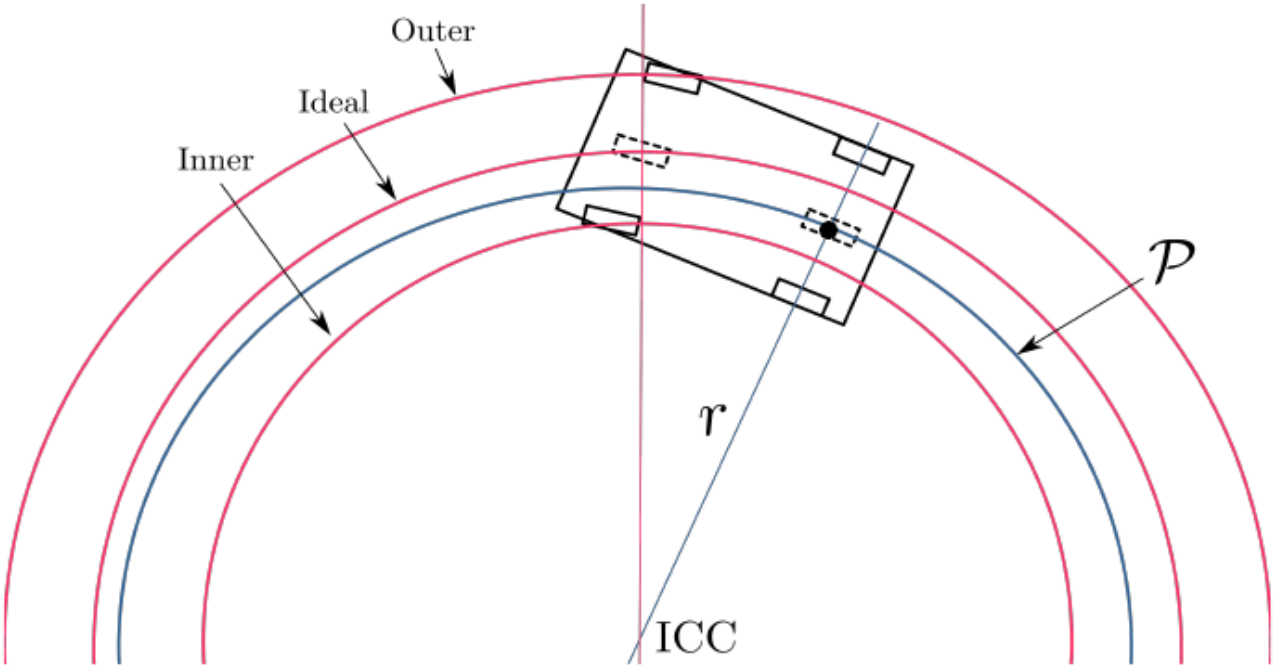
Slipping during Turning



Ackermann Model

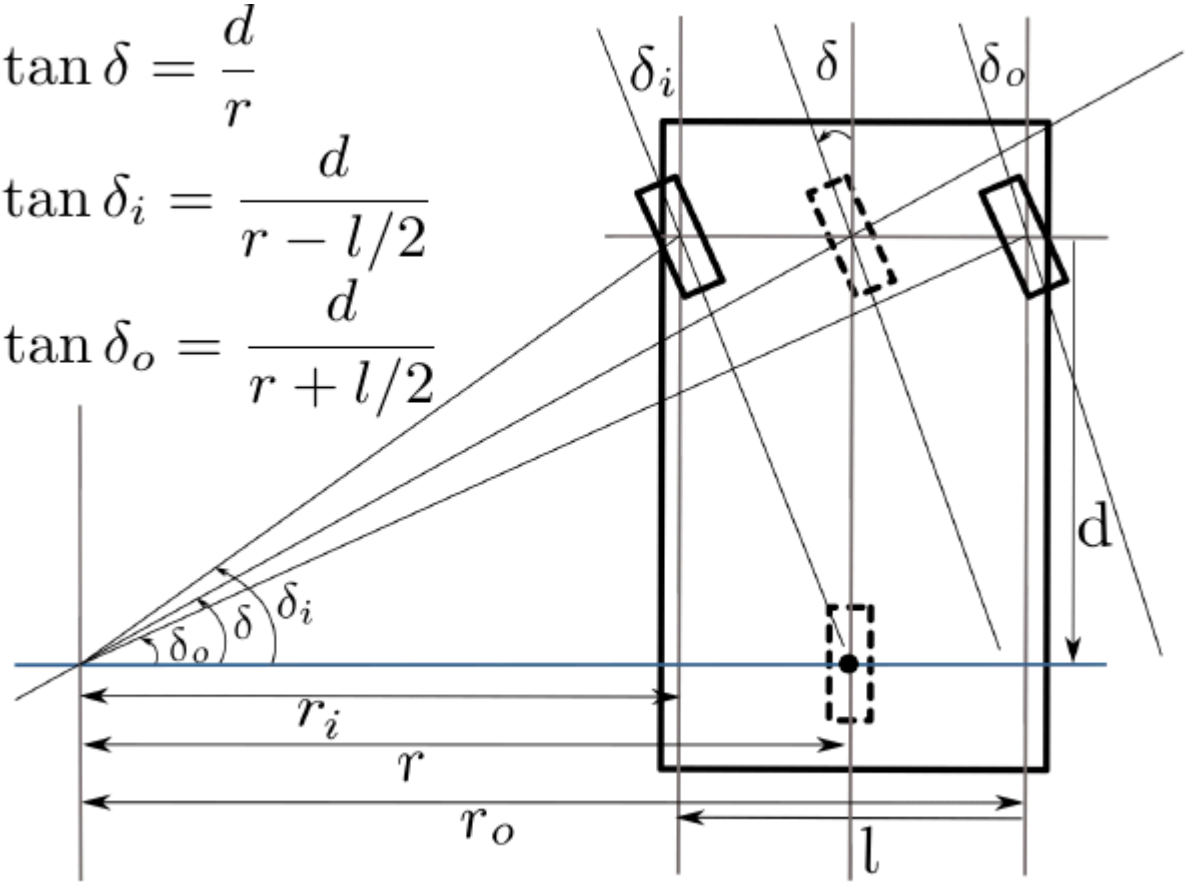


Ackermann Steering Angles

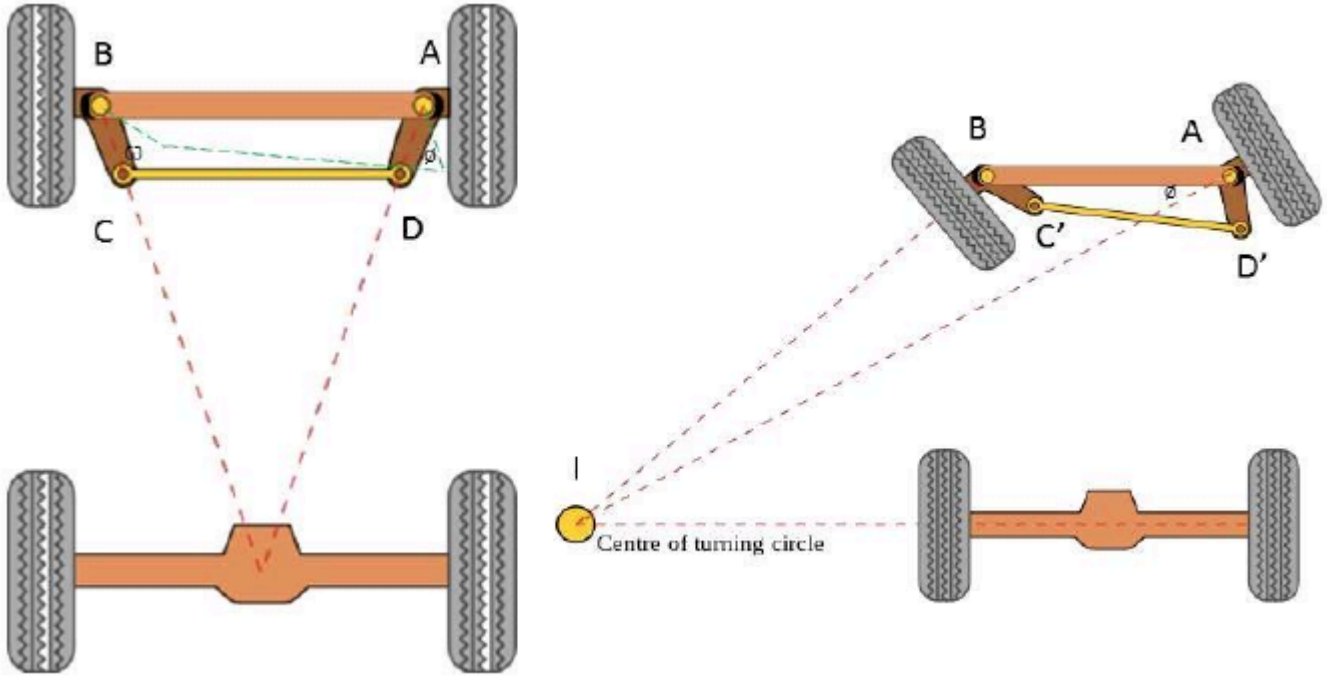
$$\tan\delta = \frac{d}{r}$$

$$\tan\delta_i = \frac{d}{r - l/2}$$

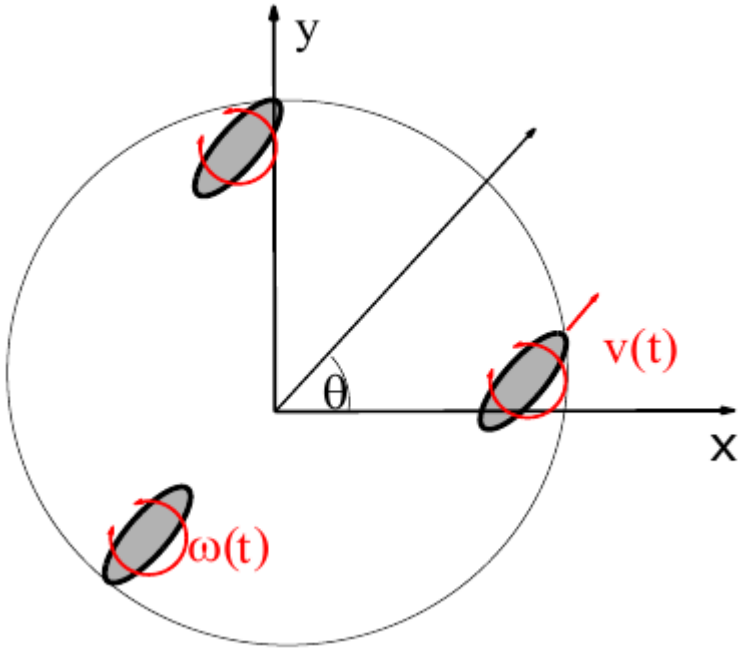
$$\tan\delta_o = \frac{d}{r + l/2}$$



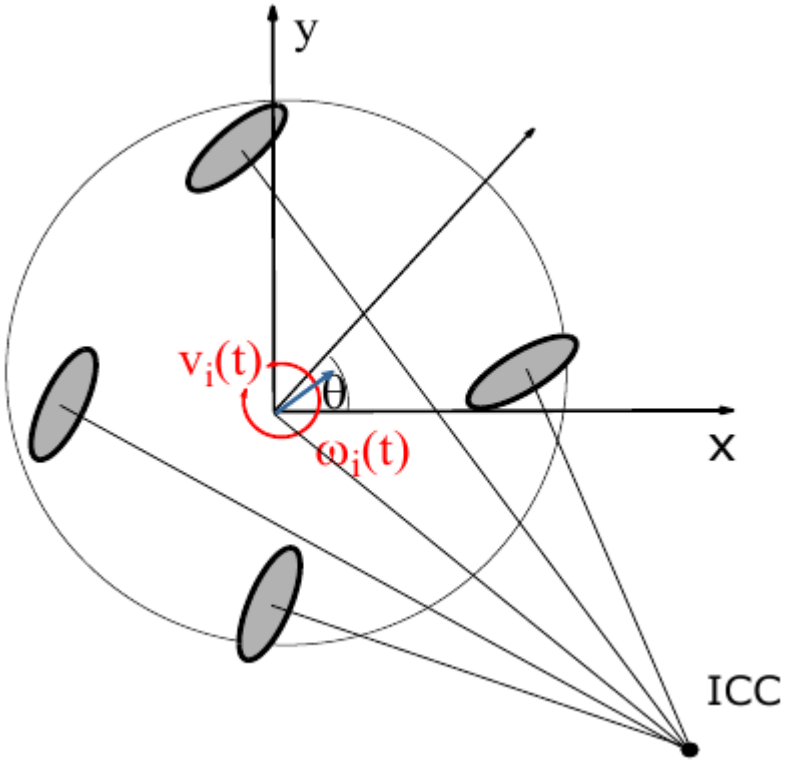
Building Ackermann Drive



Synchronous Drive



XR4000 Drive



For both:

$$x_t = \int_0^t v(t') \cos \theta_{t'} dt'$$
$$y_t = \int_0^t v(t') \sin \theta_{t'} dt'$$
$$\theta_t = \int_0^t \omega(t') dt'$$

(Non-)Holonomic Constraints

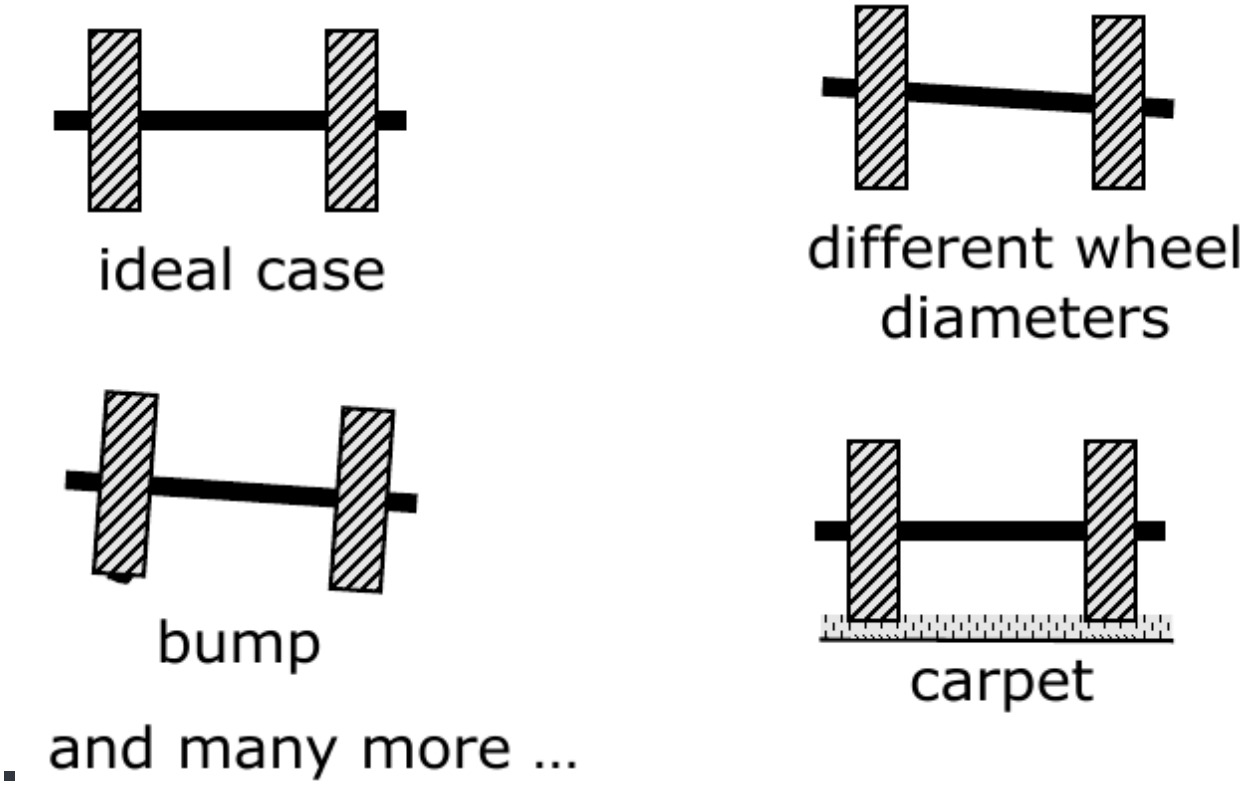
- **Holonomic constraints** reduce configuration space
  - e.g. train on tracks
- **Non-Holonomic constraints limit** the **possible movements** of the robots
  - Control space limitation w.r.t. current configuration
  - Differential & synchro-drive move on a circular trajectory and cannot execute a sideways motion
- **No non-holonomic constraints**:
  - Mecanum-wheeled robots can move sidesways
    - Mecanum-wheel: 4 independent wheels
      - Velocity is the mean velocity over all wheels

Dead Reckoning and Odometry

- Estimating motion upon controls/wheel encoder readings
- Integrated over time

Typical Motion Models

- 2 common types:
  - **Velocity-based (Dead reckoning)**
    - Pose calculation from (velocities and elapsed time)
    - Dead Reckoning from "deduced reckoning"
      - Historically used to log position of ships
    - Typically Wheel Encoders (visual black and white -> 0/5V)
  - **Odometry-based**
    - Pose calculation from **wheel encoders** values



Motion Estimation is Noisy!

Summary

- Different types of drives for wheeled robots
- Math for describing the motion of basic drives given speed of wheels
- Differential and Ackermann and 2 popular drives for wheeled platforms
- Non-Holonomic/holonomic constraints
- Odometry and dead reckoning